CSE 423 Quiz -2 | Set: A

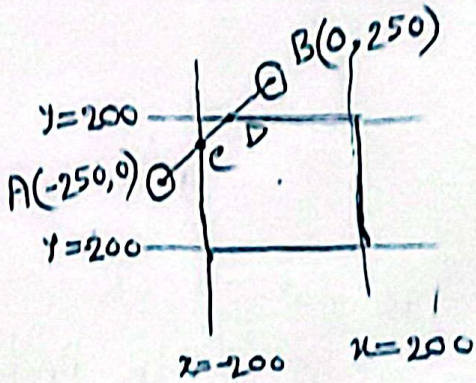
Mark: 20+6, Time: 20 minutes

Name: ID: Section:

1. Assign 4-bit outcode to the following 9 regions- 4

1001	1000	1010
0001	0000	0010
0101	0100	0110

2. Given (-200,-200) to (200,200) be the clip region. Find out whether the line (-250,0) to (0,250) is accepted/rejected/partial using Cohen Sutherland algorithm. If partially accepted find out the accepted line segment. 8



1st Iteration:

A = 0001 ≠ 0000
B = 1000 ≠ 0000
} ① can't accept

AND = 0000 ≠ Non Zero → ⑪ can't Reject

⑭ Select Non Zero (A = 0001)

AB intersect with left edge at C. AB = AC + CB (AC: outside discarded)

left edge ⇒ x = -200

AB ⇒ $\frac{y-0}{x+250} = \frac{0-250}{-250-0}$

} solving gives C = (-200, 50)

2nd Iteration:

C = 0000
B = 1000
Both zero? No } ① can't accept

AND = 0000 ≠ Non Zero → ⑪ can't reject

⑬ Select Non Zero (B = 1000)

CB intersects with top edge at D. CB = CD + DB (DB: outside discarded)

top ⇒ y = 200

AB/CB ⇒ $\frac{y-0}{x+250} = \frac{0-250}{-250-0}$

solving gives D = (-50, 200)

3rd Iteration:

C = 0000
D = 0000
Both Zero

∴ Accepted

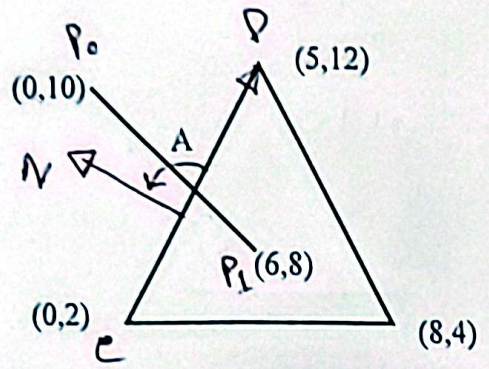
C(-200, 50), D(-50, 200)

3. Fine out the parametric equation, P(t) of the line segment (1,3) to (8,7). Then using that equation, find out the middle point of that line segment. 4

$$P(t) = P_0 + (P_1 - P_0)t = (1,3) + \{(8,7) - (1,3)\}t$$
$$= (1+7t, 3+4t)$$

mid point $\Rightarrow t = 0.5 \Rightarrow P(0.5) = (1+7\times0.5, 3+4\times0.5)$

4. Using Cyrus Beck algorithm's approach, find out the intersection point A of the following triangle- 8



$$\overrightarrow{CD} = \vec{D} - \vec{C} = (5,12) - (0,2) = (5,10)$$

Normal on $\overrightarrow{CD} = (-10,5)$ ← $(-y,x)$ → counter clockwise

$$t = \frac{N_i \cdot [P_0 - P_{Ei}]}{-N_i \cdot [P_1 - P_0]}$$

P_{Ei} = anypoint on line CD (can use C/D)

$$= \frac{(-10,5)\cdot[(0,10)-(0,2)]}{-(-10,5)\cdot[(6,8)-(0,10)]}$$

$$= \frac{40}{70} = \frac{4}{7}$$

$$P(t) = P_0 + (P_1 - P_0)t$$

$$P\left(\frac{4}{7}\right) = (0,10) + [(6,8)-(0,10)]\frac{4}{7}$$

$$= \boxed{(3.42, 8.85)}$$

5. True/False 2

a. Cohen Sutherland algorithm works for any kind of polygonal clip region. F

b. Cyrus Beck algorithm works for only convex polygon. T

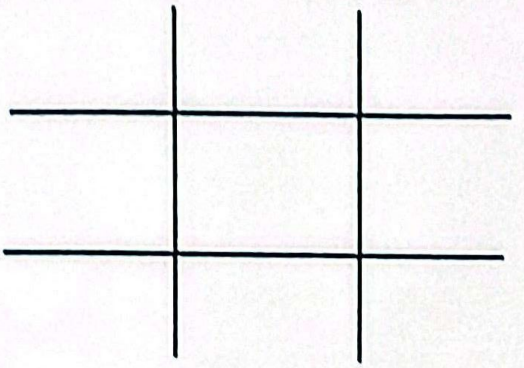
CSE 423 Quiz -2

Set: B

Mark: 20+6, Time: 20 minutes

Name: ID: Section:

1. Assign 4-bit outcode to the following 9 regions- 4



2. Given (-200,-200) to (200,200) be the clip region. Find out t values of the line (-250,0) to (0,250) and specify whether they are entering or leaving using Cyrus Beck algorithm. Also find the line segment within the clip window. 8

$x_{min} = -200$, $x_{max} = 200$, $y_{min} = -200$, $y_{max} = 200$

$P_0 = (-250, 0)$, $P_1 = (0, 250)$

$x_0 \; y_0$ $x_1 \; y_1$

$D = P_1 - P_0 = (250, 250)$

$t_E = 0$ $t_L = 1$

	Ni	Ni.D	t	PE/PL	t_E	t_L
left	(-1,0)	-250	$t = \frac{250-200}{250} = 0.2$	PE	0.2	1
Right	(1,0)	250	$t = \frac{250+200}{250} = 1.8$	PL	0.2	1
Bottom	(0,-1)	-250	$t = \frac{-200}{250} = -0.8$	PE	0.2	1
Top	(0,1)	250	$t = \frac{200}{250} = 0.8$	PL	0.2	0.8

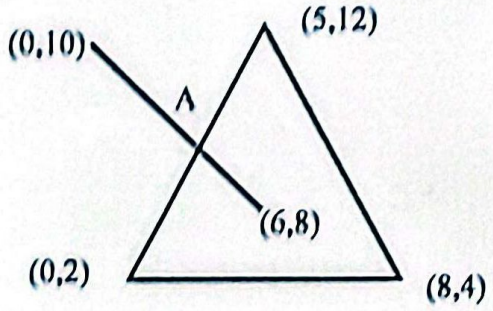
$P(0.2) = P_0 + (P_1 - P_0) \times 0.2 = (-200, 50)$

$P(0.8) = P_0 + (P_1 - P_0) \times 0.8 = (-50, 200)$

} clipped segment

3. Fine out the parametric equation, P(t) of the line segment (1,6) to (-3,8). Then using that equation, find out the middle point of that line segment. 4

4. Using Cyrus Beck algorithm's approach, find out the intersection point A of the following triangle- 8



5. True/False 2
   a. Cohen Sutherland algorithm works only for rectangle clip region. T
   b. Cyrus Beck algorithm works for only concave polygon. F

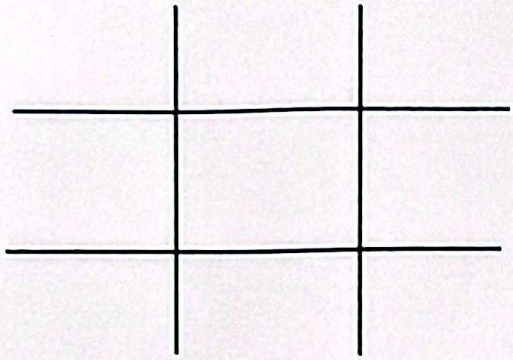
CSE 423 Quiz -2

Set: C

Mark: 20+6, Time: 20 minutes

Name: ID: Section:

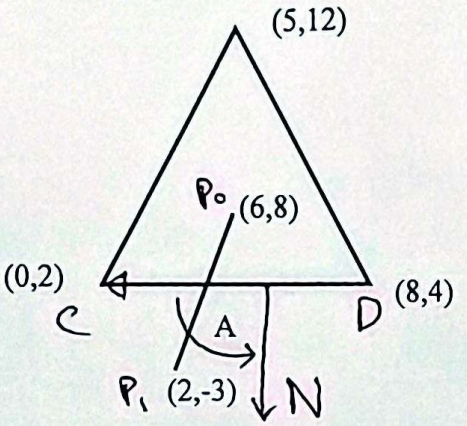
1. Assign 4-bit outcode to the following 9 regions- 4



2. Given (-200,-200) to (200,200) be the clip region. Find out whether the line (-250,0) to (0, -250) is accepted/rejected/partial using Cohen Sutherland algorithm. If partially accepted find out the accepted line segment. 8

3. Fine out the parametric equation, P(t) of the line segment (1,3) to (8,7). Then using that equation, find out the middle point of that line segment. 4

4. Using Cyrus Beck algorithm's approach, find out the intersection point A of the following triangle- 8



$\overrightarrow{DC} = \vec{C} - \vec{D} = (0,2) - (8,4) = (-8,-2)$

Normal on $\overrightarrow{DC}$ = $(2,-8)$ ← $(-y,x)$ → counter clockwise

$$t = \frac{N_i \cdot [P_0 - P_{Ei}]}{-N_i \cdot [P_1 - P_0]}$$

P_{Ei} = any point on line CD

can use C/D

$$= \frac{(2,-8) \cdot [(6,8) - (0,2)]}{-(2,-8) \cdot [(2,-3) - (6,8)]}$$

$$= \frac{-36}{-80} = 0.45$$

$P(t) = P_0 + (P_1 - P_0)t$

$P(0.45) = (6,8) + [(2,-3) - (6,8)] \times 0.45$

$= \boxed{(4.2, 3.05)}$

5. True/False 2

a. Cohen Sutherland algorithm works for any kind of polygonal clip region. F

b. Cyrus Beck algorithm works for only convex polygon. T

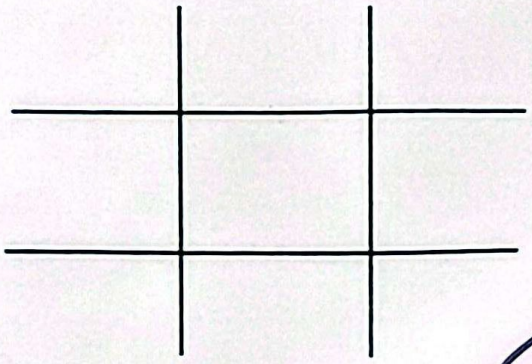
CSE 423 Quiz -2

Set: D

Mark: 20+6, Time: 20 minutes

Name: ID: Section:

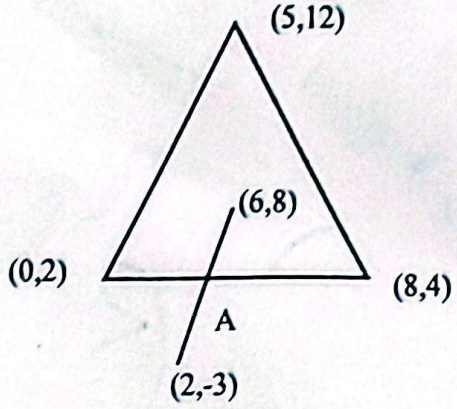
1. Assign 4-bit outcode to the following 9 regions-



2. Given (-200,-200) to (200,200) be the clip region. Find out t values of the line (-250,0) to (0,250) with clip boundaries and specify whether they are entering or leaving using Cyrus Beck algorithm. Also find the line segment within the clip window. 8

3. Fine out the parametric equation, P(t) of the line segment (1,6) to (-3,8). Then using that equation, find out the middle point of that line segment. 4

4. Using Cyrus Beck algorithm's approach, find out the intersection point A of the following triangle- 8



5. True/False 2
   a. Cohen Sutherland algorithm works only for rectangle clip region. T
   b. Cyrus Beck algorithm works for only concave polygon. F